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PAAT: Physical Activity Analysis Toolbox for the analysis of hip-worn raw accelerometer data in Python

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Software

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Summary

The Physical Activity Analysis Toolbox (*paat*) is a versatile Python package designed to analyze physical activity data. It is used in research and health-related fields to process raw acceleration data collected from the hip. It supports importing, cleaning, and preprocessing raw data, and includes algorithms to classify time in bed and non wear time. Furthermore, it supports estimating various physical activity levels such as moderate-to-vigorous physical activity or sedentary behavior, with customizable thresholds from multiple metrics. These estimates can be aggregated and used for further statistical analysis or be used directly for more sophisticated physical activity pattern analysis. Additionally, *paat* is extensible, allowing users to add custom algorithms or modules, and it integrates well with other data analysis tools within the Python ecosystem.

Statement of need

Physical activity is one of the strongest predictors of overall health. Its absence has been linked to various noncommunicable diseases such as cancer, cardiovascular diseases, or diabetes as well as mental diseases like depression or anxiety. Various methods exist to measure physical activity. One of these methods use accelerometers to estimate physical activity. Accelerometers are small body-worn sensors which measure acceleration over time. They have become a popular assessment tool in research and public health as they provide a reasonably cheap but still more objective alternative to surveys while simultaneously keeping the researcher and participant burden low.

Over decades, the raw acceleration as measured in ms^{-2} has been transformed into summary metrics like activity counts (Neishabouri et al., 2022). More recently, the raw acceleration itself has also gotten into the focus of method development (Van Hees et al., 2016). Accelerometers can be applied to various locations on the body, but most recent method development has focused on the wrist. One reason for this is its common use in many large-scale surveillance studies, like the UK Biobank (Doherty et al., 2017) or the American NHANES study (Troiano et al., 2014). This also lead to the development of the popular accelerometer analysis package GGIR (Migueles et al., 2019). Today, a plethora of accelerometer packages exist each fulfilling different purposes, for instance, to analyze accelerometer and light exposure data (Hammad et al., 2021) or to preprocess accelerometer data (Chan et al., 2024). However, none of these packages focuses on acceleration data collected from the hip even though the interest in hip-specific methods is increasing. New methods have been developed or validated recently (Syed et al., 2020, 2021; Weitz et al., 2025) as well as various large-scale population studies using hip-mounted accelerometers to measure physical activity (Hopstock et al., 2022; Weber et al., 2024).

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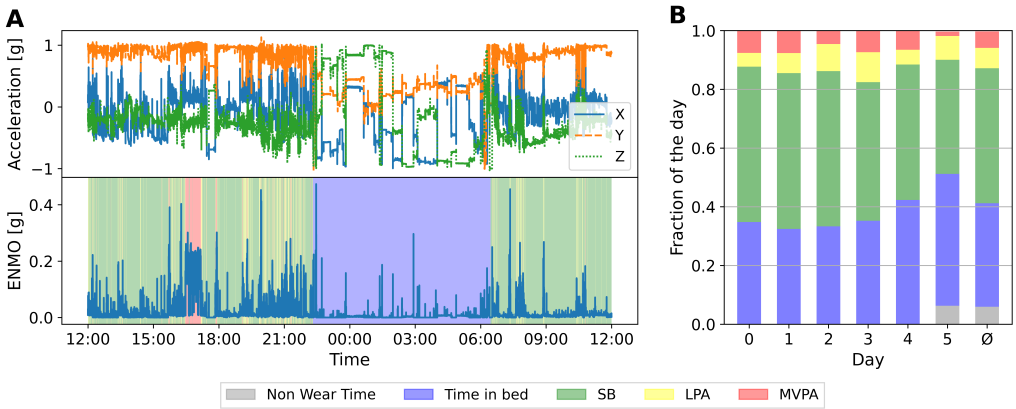

Figure 1: Visualization of the results obtained from *paat*. (A) The package can be used to load and process the raw data (upper row). The loaded data can then be annotated by a variety of methods. The implemented non-wear time and time in bed algorithm exploit raw acceleration data directly. To estimate physical activity, the raw data is reduced to the ENMO of the signal (lower row). Alternatively, also other metrics like MAD can be estimated and used for further processing. (B) Aggregated daily or average (Ø) results can be obtained and then be used for further analyzes.

For many recently developed methods, corresponding code was even available on Github, but was often difficult to use due to no existing dependency management and missing maintenance. However, easy accessibility of these methods seems particularly important to foster validation and application, especially as openly available data and standardized benchmarking procedures are scarce in the field (Palotti et al., 2019; Sadeh, 2011). Reasons for this might be the sensitive nature, but also the high costs of collecting of the data and might potentially compromise the adoption of machine learning based methods which are known to struggle to generalize. External validation of methods is an alternative to standardized benchmarking and is commonly done in this field (Palotti et al., 2019; Skovgaard et al., 2023; Syed et al., 2020) and should, thus, be facilitated.

For that reason, we collected various methods to analyze raw acceleration data from the hip in this package to facilitate validation and application of the included methods. We designed the package in a way that all methods can be run in isolation, but can also be combined to design reproducible analysis pipelines. Additionally, we designed the package to be extensible allowing users to easily add custom algorithms or use algorithms from other packages in the same pipeline by structuring it according to the respective applications (io, calibration, preprocessing, features, wear time, sleep, estimation; which are described in more detail below). By doing that, we also want to facilitate the integration into existing packages and ecosystems.

Use in research

paat has already been used in various studies. Syed et al. (2020), for instance, developed and used the general gt3x reading functionality and implemented and used the NWT algorithm from Van Hees et al. (2011) for a comparison study of different NWT algorithms. Syed et al. (2021) also used the functions to develop a new non-wear time algorithm which is now included in *paat*. Weitz et al. (2024) used the package to load and process the acceleration data to investigate the effect of accelerometer calibration on physical activity in general and MVPA in particular. Weitz et al. (2025) used the package to load and process the data in order to train a machine learning model to identify time-in-bed episodes. The developed method is now also included in this package.

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